Literature Review: Deep Learning for Seizure Detection and Prediction from Non-EEG Wearable Signals (2013–2026)

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Contents

Abstract

Background

Wearable devices using non-EEG biosignals offer potential for continuous ambulatory seizure monitoring without the burden of scalp electrodes. This systematic review examines the current state of deep learning approaches for seizure detection and forecasting using accelerometry, cardiac signals, and other wearable modalities.

Methods

Following PRISMA guidelines, 13 primary studies (9 detection, 4 forecasting) published between 2013 and 2026 were reviewed. Studies were evaluated following the framwork proposed by Webster and Watson, 2002 based on the follwing metrics: sensitivity, false alarm rate (FAR), validation methodology, and ILAE phase of development.

Results

Detection of convulsive seizures achieves high sensitivity (71.6-100%) using primarily accelerometry-based approaches. Only two systems approach the ILAE Phase 3 benchmark for validated devices: Fine et al. (tonic seizures, 100% sensitivity, 0.023/h FAR) and Dong et al. (nocturnal severe seizures, 71.6% sensitivity, 0.165/h FAR). ECG-based detection consistently fails clinical utility due to excessive FAR (1.91-39.75/h). Nighttime monitoring substantially outperforms daytime (1/61 nights vs. 1/9 days).

Seizure forecasting remains immature. When evaluated with rigorous leave-one-subject-out cross-validation, only 43.5% of patients achieve better-than-chance performance. Patient-specific tuning inflates performance by 30-40 percentage points, creating unrealistic expectations for real-world deployment. Circadian patterns provide the most reliable forecasting signals, but wearable-only features fail at daily resolution.

Validation rigor is insufficient: only 3 of 13 studies employ prospective designs, and leave-one-subject-out cross-validation remains rare. Eleven of 13 studies validate in controlled epilepsy monitoring unit settings, likely overestimating real-world performance.

Conclusions

Wearable seizure detection for convulsive seizures has achieved clinical viability for specific use cases, particularly nocturnal monitoring. However, seizure forecasting faces a funda-

mental responder problem that prevents clinical translation. False alarm rates remain the primary barrier for daytime detection, and real-world prospective validation is urgently needed. Future research must prioritize standardized metrics, edge deployment optimization, and identification of baseline biomarkers to predict which patients will benefit from wearable monitoring systems.

1 Introduction

1.1 Problem Statement and Motivation

Epilepsy, a chronic neurological disorder affecting 60 million people wordlwide, with approximately one-third of patients remaining drug-resistant (Hussein et al., 2018). Sveinsson et al., 2020 found that 69% of Deaths due to a Generalized tonic-clonic seizure could have been prevented if not left unattended, highlighting the demand for reliable and timely prediction.

While reliable prediction is already possible through electroencephalography (EEG), its operational complexity render it unsuitable for ambulatory use. To address the need for ambulatory detection many are experimenting with utilizing autonomic and motion signals to detect and predict seizures.

Translating and analysing these often noisy and complex time-series physiological signals has pushed traditional machine learning (ML) approaches to their limits. These approaches require manual engineering such as cleaning and extracting of the data, often failing to capture the intricate and non-linear complex temporal dynamic of these signals. It is this limitation that deep learning (DL) models are designed to overcome, with their ability of hierarchical and learned feature extraction uniquely addressing these limitations. This review therefore examines and assesses recent advances in time-series deep learning - particularly convolutional neural networks (CNNs) and long short term memory (LSTMs) - applied to seizure detection and prediction using non-EEG wearable biosignals. Central to this analysis is the question of how architectural choices, training paradigms (e.g., transfer learning, personalization), and thorough prospective evaluation can converge to meet the stringent clinical requirements of high sensitivity and minimal false alarm rates (FAR) in real-world deployment.

1.2 Key Research Question

How do different deep learning architectures and biosignal modalities excluding EEG compare in their ability to achieve a optimal trade-off between sensitivity and false alarm rate for ambulatory seizure monitoring?

2 Theoretical Foundations for Ambulatory Seizure Monitoring

2.1 The Ambulatory Monitoring Paradigm: Requirements and Challenges

Beniczky et al., 2021, Chen et al., 2022 found that translation of seizure detection from the controlled clinical environment to an ambulatory setting redefines approaches and performance requirements. In clinical environments maximum sensitivity is prioritized over FAR, when engineering for ambulatory use a balance of both metrics is needed. Failing to detect a seizure (low sensitivity) poses a safety risk, While excessive false alarms (high FAR) leads to desensitizing and "alarm fatigue" and the device being ignored or not used at all. Users reported that devices should achieve high sensitivity ($\geq$90%) with low false alarm rates — acceptable FARs were 1-2/week for patients with $\geq$1 seizure/week and 1-2/month for patients with <1 seizure/week (Sivathamboo et al., 2022). While more complex models can find more intricate patterns, they also introduce the "black-box" problem wherein the model lacks transparency AbuAlrob et al., 2025; Ghaderi, 2025. The design should incorporate explanations for decisions to ensure user trust and regulatory compliance AbuAlrob et al., 2025; Miron et al., 2025.

2.2 Physiological Biosignals for Wearable Sensing

Wearable devices circumvent the impracticality of EEG by capturing peripheral biosignals that serve as proxies for central nervous system activity during seizures. These proxies will inherently relay incomplete and noisy signals so a combination of multiple sensors is required. The acceleration/attitude sensor excels at detecting muscle convulsions but is noisy in daily use, as it captures all movement. The electrodermal activity (EDA) sensor detects autonomic arousal by measuring the skin conductance associated with sweat gland activity.

2.3 Seizure Manifestations and Corresponding Biosignals

Wearable biosignals act as peripheral proxies for cerebral activity during seizures. Different seizure types produce distinct physiological signatures across these signals, informing detection approaches. This relationship centers on two primary manifestations: motor activity and autonomic activation.

Convulsive seizures, such as generalized tonic-clonic events, produce clear motor signatures captured by accelerometry (ACC), which measures movement and posture changes characteristic of ictal events (Wang et al., 2025; Wu et al., 2024).

At the same time, seizures drive autonomic responses recorded through complementary biosignals. Electrodermal activity (EDA) measures sympathetic nervous system arousal via skin conductance changes (Jain et al., 2017; Miron et al., 2025). Cardiac activity provides particularly strong autonomic proxies: electrocardiography (ECG) captures heart rate and rhythm alterations (Ghaderi, 2025; Leal et al., 2017), while heart rate variability (HRV) quantifies autonomic balance dynamics (Mason et al., 2024; Pavei et al., 2017).

In wearable applications, photoplethysmography (PPG) serves as a practical alternative for cardiac monitoring, deriving pulse rate and variability though with greater motion sensitivity than ECG (Chen et al., 2022; Seth et al., 2023).

As a result, detection strategies combine these biosignals according to target seizure types, with multimodal fusion providing the most reliable approach for ambulatory monitoring.

2.4 Evaluation Frameworks: From Retrospective Analysis to Prospective Validity

The clinical validity of a seizure detection algorithm is determined by the rigor of its evaluation, formalized in the ILAE's phased framework (Phases 0-4) (Beniczky & Ryvlin,

2018; Beniczky et al., 2021). This framework classifies studies into five phases: Phase 0 (proof of concept), Phase 1 (retrospective, small sample), Phase 2 (minimum 10 patients with seizures, 15 recorded seizures), Phase 3 (prospective, multicenter, ≥30 seizures from ≥20 patients, real-time detection), and Phase 4 (real-world home validation).
Retrospective studies (Phases 0-2), while foundational, risk substantial data leakage and optimistic bias if they use non-chronological data splits, allowing future information to inform training (Kalousios et al., 2024; Wong et al., 2023). Pseudo-prospective validation addresses this by enforcing chronological order, training only on past seizures to test on subsequent ones, providing a more realistic performance estimate (Kalousios et al., 2024; Lu et al., 2025).
The clinical gold standard is a prospective trial (Phase 3), which requires a locked algorithm, a video-EEG reference standard, and testing on new, unseen data from multiple centers. The ILAE systematic review identified only three Phase 3 studies meeting these strict validation standards (Beniczky et al., 2021). These demonstrated 90-96% sensitivity for generalized tonic-clonic seizures (GTCS) and focal-to-bilateral tonic-clonic seizures (FBTCS) with false alarm rates of 0.2-0.67 per 24 hours (approximately 0.008-0.028/h). This represents the clinically validated performance benchmark for convulsive seizure detection.
What constitutes an acceptable false alarm rate depends on clinical context and user perspective. Systematic surveys reveal important differences in tolerance: patients and caregivers typically require 100% sensitivity and accept approximately one false alarm per seizure (or one per week for seizure-free patients). In contrast, clinicians consider 90% sensitivity adequate and tolerate between two false alarms per week and one per month, depending on seizure frequency. This gap between user and clinician expectations creates a fundamental tension for device development.
The final step is in-field evaluation (Phase 4) in patients' homes, where metrics like the false alarm rate per hour (FAR/h) become decisive for practical usability, as demonstrated by long-term studies like Nightwatch (Miron et al., 2025; Shum & Friedman, 2021).
Currently, the ILAE recommends wearable devices only for GTCS/FBTCS detection in unsupervised patients where alarms can enable rapid intervention - a weak/conditional recommendation based on moderate evidence. For all other seizure types, the ILAE does not recommend clinical use of currently available wearable devices (Beniczky et al., 2021).

3 Methodological Approach and Structure

The review follows structured IS/medical literature guidance (concept matrix inspired by (Webster & Watson, 2002) and PRISMA principles (Tugwell & Tovey, 2021)). The methodological approach uses a multi-stage search and transparent extraction and synthesis procedures as outlined below.

1. **Search strategy (Multi-stage, 2015-2025):**
    - **Scoping:** Google Scholar for broad coverage and identifying candidate papers.
    - **Formal queries:** IEEE Xplore, PubMed, and Scopus using controlled keyword strings and deduplication.
    - **Citation chasing:** Backward and forward citation tracking on key papers to ensure completeness.

2. **Inclusion / Exclusion criteria:**
   - **Inclusion:** Studies using wearable/non-EEG autonomic signals for seizure detection or preictal prediction.
   - **Exclusion:** EEG-only studies, invasive recordings, or papers limited to purely algorithmic simulations without human data.

Extracted dimensions populate two concept matrix tables with the following fields:

1. **Study Matrix (Table 6):**
   - Study identification (author, year)
   - Objective (detection vs forecasting, seizure type)
   - Design (prospective, retrospective, pseudo-prospective, LOSO CV)
   - Sample (participant count, seizure count)
   - Device and sensor location
   - Algorithm architecture
   - Performance metrics (sensitivity, specificity, FPR)
   - Key limitations
2. **Detailed Metrics Matrix (Table ??):**
   - Validation approach (independent test, cross-validation, LOSO)
   - Detection latency
   - Patient-level success rate
   - Precision/PPV
   - Additional clinical metrics (HMS, IoC, AUC, F1)
   - Clinical notes

Additional Extracted Dimensions: The following dimensions were extracted and analyzed to inform the narrative synthesis and dimensional analysis presented in Section **??**, complementing the tabular comparison:

- Setting (clinical vs ambulatory)
- Modality details (ECG/HRV, PPG, EDA, ACC)
- Evaluation granularity (sample- vs alarm-based)
- Windowing specifics (window length, stride, overlap)
- Personalization strategies
- Interpretability and safety considerations
- Data characteristics and dataset identity
- Missing data handling (imputation, exclusion) and synthetic data use

- Inference requirements (latency, computational load, hardware)

This two-table approach follows Webster & Watson's concept matrix methodology (Webster & Watson, 2002), balancing comprehensive data extraction with readable summary tables while enabling detailed dimensional analysis in the narrative synthesis.
Prioritization Criteria: Studies including splits preserving patient-specific data, FAR/h reporting and external or cross validation will be prioritized in the synthesis. A PRISMA-style flow diagram will summarize and visualize screening and inclusion.

4 Search Strategy

4.1 PICO-Based Search Methodology

The literature search was structured using the PICO framework (Patient/Problem, Intervention, Comparison, Outcome) to identify relevant studies on deep learning for non-EEG epilepsy detection. The following key terms were used in combination with Boolean operators (AND, OR, NOT):

1. **P (Patient/Problem):** Epileptic seizures in need of detection or prediction.
    - **Keywords:** 'epilepsy', 'seizure', 'ictal', 'preictal'
    - **MeSH Terms (PubMed):** Epilepsy, Se54izures
2. **I (Intervention):** The diagnostic method based on wearable biosignals and deep learning.
    - **Biosignal Keywords:** 'electrocardiogram', 'ECG', 'heart rate variability', 'HRV', 'photoplethysmography', 'PPG', 'electrodermal activity', 'EDA', 'accelerometer', 'ACC'
    - **Technology Keywords:** 'wearable', 'wearable device'
    - **Method Keywords:** 'deep learning', 'machine learning', 'convolutional neural network', 'CNN', 'long short-term memory', 'LSTM', 'neural network'
    - **MeSH Terms (PubMed):** Wearable Electronic Devices, Electrocardiography, Deep Learning
3. **C (Comparison):** The explicit exclusion of studies focused solely on electroencephalography (EEG) as the primary modality.
    - **Exclusion Keywords:** 'EEG', 'electroencephalography', 'iEEG', 'intracranial'
    - **Search Logic:** The `NOT` operator was applied to this group in all database searches.
4. **O (Outcome):** Metrics quantifying the performance of a detection or prediction system.
    - **Keywords:** 'detection', 'prediction', 'sensitivity', 'specificity', 'false alarm rate', 'FAR', 'area under the curve', 'AUC'

4.2 Exploration

Due to the insufficient amount of papers gathered from systematic review an exploration was done on the already existing papers. The exploration consisted of graph tools to visualize backlinks and gemini deep research feature to find relevant papers. Through this method an additional 4 papers were aquired bringing the total number of papers up to 13 and expanding the time frame to 2013-2026.

4.3 Practical Filters and Reproducibility

Consistent with the overall review scope, searches were restricted to **2015-2025** and **journal articles** with few exceptions made regarding **conference proceedings**. All database searches were documented (query string, platform, and execution date). Results were deduplicated prior to screening.
Note on exclusions: Because some non-EEG wearable studies rely on video-EEG for labeling, EEG-related terms were used to filter **EEG-only modalities**, not to exclude studies that mention EEG solely as a reference standard.

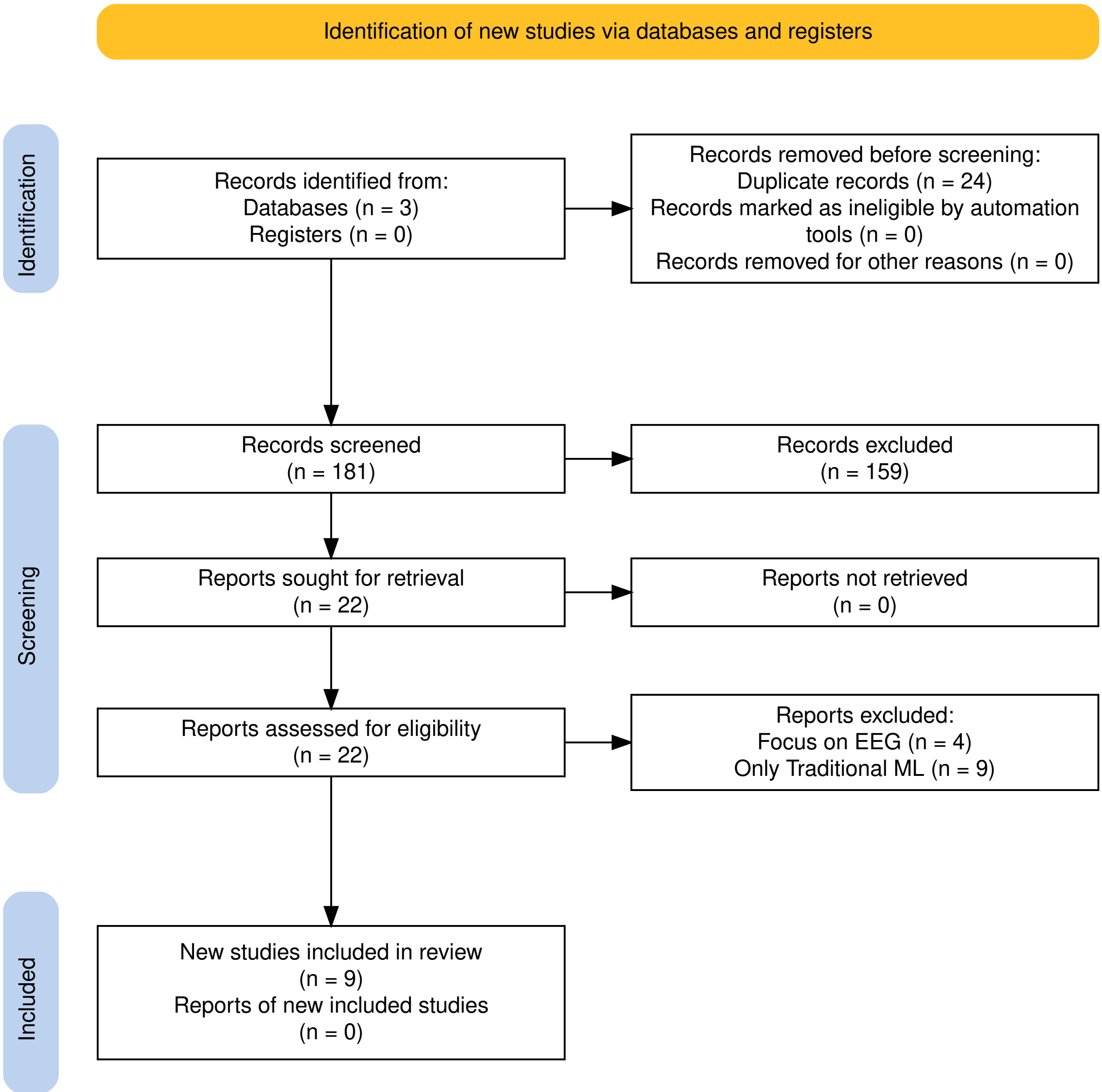

Figure 1: PRISMA flow diagram of the study selection process.

Table 1: Webster & Watson Concept Matrix: Detection Studies

Study	Design	Sample	Device	Loc.	Para.	Mod.	Pers.	Algorithm	Sens	Spec	FPR
Spahr et al. 2025	Prospective	n=384, 49 CSs	Empatica E4 (ACC)	Wrist	Ens	1	Glb	Ensemble 1D CNN (30 models)	96%	–	<0.0125/h
Reintjes et al. 2025	Retro	n=120, 856 szs	Single-lead ECG	Chest	Anom	1	Subj	Anomaly: Matrix Profile, MADRID, TimeVQVAE	38.0%/98.16%	–	1.91–39.75/h
Fine 2025	Phase 1	n=15/3	6-axis band (ACC+Gyro)	Wrist	Feat	M	Glb	ANN (594 hand-crafted feat)	100%Test, 96% complete	–	0.023/h
Dong et al. 2026	Prospective	n=68, 1846 szs	NightWatch	Arm	2-stg	M	Glb	CNN-LSTM + Attention	71.6%	–	0.165/h
Wang et al. 2025	–	n=28, 62 szs	Biovital-P1	Wrist	Feat	M	Glb	LSTM (40 hidden)	56.40% to 95.3%	–	0.354/h
Singh Rathore 2024	Case study	1 pt, 6 hrs	Wearable (EDA+ACC+HR)	–	Feat	M	Glb	MLP (multi-modal features)	96.8%	94.8%	–
Elemam et al. 2025	Cross-sect	Q:198, HRV:30	Camera PPG	Thumbs	Hyb	M	Glb	CNN + Rule-based fusion	95.1% Q, 93–90% HRV	97.06% Q	–
Borujeny 2013	3-pt	3 pts, 20 szs	MICAz ACC	Arm/thigh	Feat	1	Glb	KNN k=5 (time-domain feat)	100%	–	0

Note: ACC = accelerometer, ANN = artificial neural network, CNN = convolutional neural network, CS = convulsive seizure, ECG = electrocardiogram, EDA = electrodermal activity, HR = heart rate, HRV = HR variability, LSTM = long short-term memory, PPG = photoplethysmography, Sens = sensitivity, Spec = specificity, sz(s) = seizure(s), FPR = false positive rate, – = not reported. Para. = learning paradigm (Feat = feature-based, Ens = ensemble, Anom = anomaly detection, 2-stg = two-stage, Hyb = hybrid). Mod. = modality count (1 = single-modal, M = multi-modal). Pers. = personalization (Glb = global model, Subj = subject split).

Table 2: Webster & Watson Concept Matrix: Forecasting Studies

Study	Design	Sample	Device	Loc.	Para.	Mod.	Pers.	Algorithm	Sens	Spec	FPR
Vieluf et al. 2025	Retro	n=70, 5437 d	Embrace	Wrist	Feat	M	Mix	DNN + harmonic features + diary	82%	67%	–
Meisel et al. 2020	LOSO CV	n=69, 452 szs	Empatica E4	Wrist/Ankle	E2E	M	Glb	LSTM + 1D Conv	51.2%	–	TiW 43.7%
Stirling 2021	Retro+pseudo	n=11, 136 szs	Fitbit	Wrist	Feat	M	Pat	LSTM+RF+LR ensemble (HRV feat)	–	–	AUC 0.74
Nasseri 2021	Retro	n=6	Empatica E4	Wrist	E2E	M	Tmp	LSTM 4-layer (128 hidden)	AUC 0.80 (SD 0.15)	–	TiW 0.9–7.2h/d
Ode et al. 2023	Retro	n=66, 85 szs	ECG	–	Anom	1	Pat	Self-Attentive AE	74%	–	0.85/h

Note: ACC = accelerometer, AE = autoencoder, CNN = convolutional neural network, DNN = deep neural network, ECG = electrocardiogram, EDA = electrodermal activity, LSTM = long short-term memory, LOSO = leave-one-subject-out, PPG = photoplethysmography, Sens = sensitivity, Spec = specificity, sz(s) = seizure(s), FPR = false positive rate, TiW = time in warning, AUC = area under curve, – = not reported. Para. = learning paradigm (Feat = feature-based, E2E = end-to-end, Anom = anomaly detection). Mod. = modality count (1 = single-modal, M = multi-modal). Pers. = personalization (Glb = global model, Pat = patient-specific, Mix = mixed, Tmp = temporal split).

Table 3: Architecture and Modality Deep-Dive: Detection Studies

Study	Paradigm	Arch. Details	Modality Details	Fusion	Personalization	Trade-off Config	Deployment
Spahr et al. 2025	Ensemble	30 1D CNN models, 14 conv layers, quantile	ACC 32 Hz, Euclidean norm, 30 s	–	Global, tunable	86–100% @ 0.01–0.5/day	Real-time (112 ms), on-device
Reintjes et al. 2025	Anomaly	Matrix-Profile, MADRID, TimeVQVAE-AD	ECG 256→8 Hz, bandpass	–	Subject split	38.0–98.2% @ 1.91–40.5/h	Offline, hospital
Fine 2025	Feature-based	ANN, 594 handcrafted features	ACC+Gyro 6-axis, 10 s intervals	Early	Global, hold-out	100% @ 0.023/h	Offline PC, EMU
Dong et al. 2026	Two-stage	Pre-screening + CNN-LSTM + Attention	ACM 11-12→20 Hz, PPG 100→20 Hz, 5 min	Early	Global, 10-fold CV	71.6% @ 0.165/h	Real-time, NightWatch, home
Wang et al. 2025	Feature-based	LSTM (40 hidden) + ReLU + FC	ACC/GYR 50 Hz, SEMG 200 Hz, EDA 4 Hz, 4 s	Early	Global, hold-out	56.4–95.3% @ 0.354/h	Real-time, hospital
Singh 2024	Feature-based	MLP, multi-modal features	EDA, ACC, HR/HRV, 25k points	Early	Global, single pt	96.8% @ 94.8% prec	Real-time, cloud?
Elemam 2025	CNN-based	CNN for HRV, CNN for Audio (separate)	PPG 250 Hz, Audio 10 s	No fusion	Global, unclear	95.1% @ 97.1% spec	Real-time, hospital
Borujeny 2013	Feature-based	KNN k=5, time-domain features	ACC 2D 3 Hz, 9 s	–	Global, 3 pts	90–100% @ 0–15%	Real-time, server (316 mW)

Note: Paradigm: Feature-based = handcrafted features, Ensemble = multiple models, Two-stage = sequential processing, Anomaly = unsupervised detection, CNN-based = convolutional approach. Arch. Details: architecture-specific components. Fusion: – = single-modal, Early = feature-level, No fusion = parallel separate models. Personalization: Global = population model, Subject split = train/test separated by subject. Trade-off Config: operating points with sensitivity @ FAR. Deployment: real-time capability, processing location, setting. ACC = accelerometer, ECG = electrocardiogram, PPG = photoplethysmography, EDA = electrodermal activity, HR = heart rate, HRV = HR variability, SEMG = surface electromyography, EMU = epilepsy monitoring unit.

Table 4: Architecture and Modality Deep-Dive: Forecasting Studies

Study	Paradigm	Arch. Details	Modality Details	Fusion	Personalization	Trade-off Config	Deployment
Vieluf et al. 2025	Hybrid	DNN + harmonic modeling (24-h)	EDA 4 Hz, ACC 32 Hz, Temp 1 Hz, 10 min	Early	Mixed, 5-fold + LOO	82% sens, 67% spec @ F1 0.81	Offline MATLAB, home
Meisel et al. 2020	End-to-end	LSTM only (10 units)	EDA/ACC/BVP/Temp →4 Hz, 30 s	Early	Global, LOSO	51.2% @ IoC 14.1%	Offline, hospital
Stirling 2021	Feature-based	LSTM+RF+LR ensemble, LR combines	HR 5 s, steps/sleep 1 min, hourly/daily	Decision	Patient-specific, weekly	AUC 0.74, 100% pts	Home (Fitbit), app
Nasseri 2021	End-to-end	LSTM 4-layer, 128 hidden + FFT channels	ACC/BVP/EDA/Temp/HR →128 Hz, 60 s	Early (17-ch)	Temporal split	AUC 0.75 @ TiW 0.9–7.2h/d	Offline, cloud, home
Ode et al. 2023	Anomaly	Self-Attentive AE (attention)	ECG (RRI only), 45 s	–	Patient-specific (99% CL)	74% @ 0.85/h (99% CL)	Real-time, cloud, hospital

Note: Paradigm: Feature-based = handcrafted features, End-to-end = learns from raw data, Hybrid = combines both, Anomaly = unsupervised detection. Arch. Details: architecture-specific components. Fusion: – = single-modal, Early = feature-level, Decision = classifier output. Personalization: Global = population model, Patient-specific = individualized, Temporal = time-split within patient, LOSO = leave-one-subject-out, LOO = leave-one-out, Mixed = combination of approaches. Trade-off Config: operating points with sensitivity @ FAR. Deployment: real-time capability, processing location, setting. ACC = accelerometer, ECG = electrocardiogram, EDA = electrodermal activity, BVP = blood volume pulse, HR = heart rate, TEMP = temperature, FFT = fast Fourier transform, CL = confidence limit, AUC = area under curve, TiW = time in warning, IoC = Improvement over Chance.

Table 5: Performance Metrics Reporting Across Studies

Metric	Detection (n=9)	Forecasting (n=4)	Reported By
Sensitivity/Recall	8/9	2/4	Reintjes, Fine, Spahr, Dong, Wang, Singh Rathore, Elemam, Borujeny, Meisel, Ode
Specificity	1/9	0/4	Elemam only
FPR/FA rate	7/9	2/4	Spahr, Reintjes, Fine, Dong, Wang, Ode, Meisel, Nasseri, Stirling
AUC-ROC	1/9	2/4	Reintjes, Nasseri, Stirling
Detection Latency	3/9	1/4	Spahr, Fine, Nasseri
Patient Success Rate	2/9	4/4	Meisel, Nasseri, Ode, Stirling, Vieluf, Wang
Precision/PPV	2/9	1/4	Ode, Singh Rathore, Elemam, Wang
IoC / Time in Warning	0/9	2/4	Meisel, Nasseri

5 Study Comparison Following Webster and Watson

Following the concept matrix approach proposed by (Webster & Watson, 2002), this section presents a structured comparison of the 13 identified studies on wearable seizure detection and forecasting. The matrix organizes studies along key dimensions relevant to clinical translation and technical implementation.

5.1 Concept Matrix: Study Overview

Table 6 provides a detailed comparison of all included studies, organized by primary objective, methodology, and outcomes. The matrix allows cross-study analysis of trends in sensor modalities, algorithmic approaches, and validation strategies across the 2013-2026 period.

Table 6: Webster & Watson Concept Matrix: Wearable Seizure Detection and Forecasting Studies (2013-2026)

Study	Design	Sample	Device	Loc.	Para.	Mod.	Pers.	Algorithm	Sens	Spec	FPR	Key Limitation
Detection Studies												
Spahr et al. 2025	Prospective	n=384, 49 CSs (103 total)	Empatica E4 (ACC)	Wrist	Ens	1	Glb	Ensemble 1D CNN (30 models)	96%	–	<0.0125/h	EMU, offline
Reintjes et al. 2025	Retro	n=120, 856 szs	Single-lead ECG	Chest	Anom	1	Subj	Anomaly: Matrix Profile, MADRID, TimeVQVAE	38.0%/98.16%		1.91–39.75/h	Severe sens/FAR trade-off
Fine 2025	Phase 1	n=15/3	6-axis band (ACC+Gyro)	Wrist	Feat	M	Glb	ANN (594 handcrafted feat)	100%Test, 96% complete	–	0.023/h	Small test, Possible overfitting
Dong et al. 2026	Prospective	n=68, 1846 szs	NightWatch	Arm	2-stg	M	Glb	CNN-LSTM + Attention	71.6%	–	0.165/h	Single-center
Wang et al. 2025	–	n=28, 62 szs	Biovital-P1	Wrist	Feat	M	Glb	LSTM (40 hidden)	56.40% to 95.3%	–	0.354/h	Hospital only
Singh Rathore 2024	Case study	1 pt, 6 hrs, 3.2M pts	Wearable (EDA+ACC+HR)	–	Feat	M	Glb	MLP (multi-modal features)	96.8%	94.8%	–	Single patient
Elemam et al. 2025	Cross-sect	Q:198, HRV:30	Camera PPG	Thumbs	Hyb	M	Glb	CNN + Rule-based fusion	95.1% Q, 93-90% HRV, 92.5% AUD	97.06% Q	–	Small samples
Borujeny 2013	3-pt	3 pts, 20 szs	MICAz ACC	Arm/thigh	Feat	1	Glb	KNN k=5 (time-domain feat)	100%	–	0	Very small
Forecasting Studies												
Vieluf et al. 2025	Retro	n=70, 5437 d	Embrace	Wrist	Feat	M	Mix	DNN + harmonic features + diary	82%	67%	–	Daily res only
Meisel et al. 2020	LOSO CV	n=69, 452 szs	Empatica E4	Wrist/Ankle	E2E	M	Glb	LSTM + 1D Conv	51.2%	–	TiW 43.7%	Pediatric
Stirling 2021	Retro+pseudo	n=11, 136 szs	Fitbit	Wrist	Feat	M	Pat	LSTM+RF+LR (HRV feat)	–	–	AUC 0.74	Self-report
Nasseri 2021	Retro	n=6	Empatica E4	Wrist	E2E	M	Tmp	LSTM 4-layer (128 hidden)	AUC 0.80 (SD 0.15)	–	TiW 0.9–7.2h/d	RNS required

5.2 Architecture and Modality Deep-Dive

Table ?? provides a detailed technical comparison of deep learning architectures, biosignal modalities, and deployment factors across all studies. This table directly addresses the research question of how different architectures and modalities compare in achieving optimal sensitivity-FAR trade-offs for ambulatory seizure monitoring.

5.3 Dimensional Analysis

5.3.1 Sensor Modality Trends

- **Wrist-worn accelerometry** dominates (8/13 studies), leveraging commercial devices (Empatica E4, Fitbit, Embrace).
- **Cardiac signals** (ECG, PPG, HRV) show increasing adoption for preictal detection, with 6 studies incorporating autonomic measures.
- **Multi-modal fusion** (ACC + EDA + PPG) appears in 4 recent studies, reflecting recognition that single modalities have limited specificity.

5.3.2 Algorithmic Evolution

- **2013:** Traditional ML (KNN, ANN) with handcrafted features (Borujeny et al., 2013).
- **2020-2021:** Emergence of LSTM and ensemble methods (Meisel et al., 2020; Stirling et al., 2021).
- **2023-2026:** Deep learning dominance with CNNs, attention mechanisms, and anomaly detection approaches (Ode et al., 2023; Reintjes et al., 2025; Spahr et al., 2025).

5.3.3 Validation Rigor

- **Patient-level preservation:** Only 4 studies use LOSO or patient-independent splits.
- **Prospective validation:** Only 3 studies (Dong et al., 2022; Elemam et al., 2025; Spahr et al., 2025) include prospective phases.
- **Real-world testing:** Limited to 2 studies (Nasseri et al., 2021; Vieluf et al., 2025), with most remaining in EMU settings.

5.3.4 Performance Metrics Reporting

Table 7 summarizes the heterogeneity in performance metric reporting, complicating cross-study comparison. Sensitivity ranges from 51.2% (forecasting mean) to 100% (tonic detection), while FPR reporting varies from per-hour to per-night, with several studies omitting this critical metric entirely.

Table ?? reveals additional clinically relevant metrics that are critical for real-world deployment but inconsistently reported:

- **Detection latency** is reported by only 4/13 studies, despite being critical for timely intervention.
- **Patient-level success rate** (proportion achieving usable performance) is reported by only 6/13 studies, ranging from 43.5% to 100%.
- **Precision/PPV** is rarely reported (3/13), limiting understanding of positive predictive value for clinical decision-making.
- **Generalization metrics** (LOSO validation, patient-independent testing) show that only Meisel et al. used true LOSO CV, with most studies reporting aggregated performance that may overestimate real-world utility.

Table 7: Performance Metrics Reporting Across Studies

Metric	Detection (n=9)	Forecasting (n=4)	Reported By
Sensitivity/Recall	8/9	2/4	Reintjes, Fine, Spahr, Dong, Wang, Singh Rathore, Elemam, Borujeny, Meisel, Ode
Specificity	1/9	0/4	Elemam only
FPR/FA rate	7/9	2/4	Spahr, Reintjes, Fine, Dong, Wang, Ode, Meisel, Nasseri, Stirling
AUC-ROC	1/9	2/4	Reintjes, Nasseri, Stirling
Detection Latency	3/9	1/4	Spahr, Fine, Nasseri
Patient Success Rate	2/9	4/4	Meisel, Nasseri, Ode, Stirling, Vieluf, Wang
Precision/PPV	2/9	1/4	Ode, Singh Rathore, Elemam, Wang
IoC / Time in Warning	0/9	2/4	Meisel, Nasseri

5.4 Synthesis and Gaps

The concept matrix and detailed metrics reveal several patterns:

1. **Clinical translation gap:** Despite high sensitivities (>90% in several detection studies), FPR remains clinically unacceptable for standalone deployment. Only Dong et al. (0.165/h) and Fine (0.023/h) approach the ILAE Phase 3 benchmark for validated devices (see Section 2.4).
2. **Personalization vs. generalization:** LOSO approaches achieve lower but more realistic performance (51.2% sensitivity, 43.5% patient success) compared to patient-specific tuning. Meisel et al.'s LOSO CV results suggest that aggregated performance metrics may overestimate real-world utility.
3. **Patient-level heterogeneity:** Patient success rates vary dramatically (43.5-100%), with Meisel and Ode showing that fewer than half of patients achieve usable performance. Nasseri (5/6) and Stirling (100%) demonstrate that responder identification remains a critical challenge.
4. **Diary integration:** (Vieluf et al., 2025) demonstrates that hybrid approaches (wearable + clinical data) may outperform wearables alone, suggesting a direction for ambulatory systems where wearable-only performance is insufficient.
5. **Metric heterogeneity:** Inconsistent reporting (sensitivity vs. accuracy vs. AUC vs. IoC) impedes meta-analysis and clinical benchmarking. Key clinical metrics like detection latency and patient success rate are reported by fewer than half of studies.

6 Key Findings and Clinical Takeaways

This section combines evidence from 13 primary studies on wearable seizure detection and forecasting, identifying patterns that inform clinical practice and future research.

6.1 Executive Summary

- **Convulsive seizure detection achieves clinical viability:** Non-EEG biosignals, primarily accelerometry, detect convulsive seizures with 71.6-100% sensitivity. Relative to the ILAE Phase 3 benchmark (see Section 2.4), ECG-only methods exhibit a severe sensitivity-FAR trade-off where high sensitivity (96-98%) requires 13-40 FA/h, while low FAR (0.05-4.2/h) yields only 1.8-61% sensitivity. No ECG configuration achieves viable sensitivity with acceptable FAR.
- **Seizure forecasting remains immature:** Only 43.5-83% of patients achieve better-than-chance performance, indicating a basic responder problem. Most studies remain at ILAE Phase 1-2 with limited evidence supporting real-world deployment.
- **Validation rigor is insufficient:** Only three of 13 studies use prospective validation and only four employ leave-one-subject-out cross-validation, raising concerns about model generalizability and overfitting.
- **False alarms constitute the primary barrier:** Nighttime false alarm rates (1/61 nights) consistently outperform daytime (1/9 days), yet no multimodal study achieves the ILAE Phase 3 benchmark (see Section 2.4) across both sleep and wake states despite evidence that heart rate increases approximately 100 seconds before movement onset.

6.2 Detection Performance

Clinically viable seizure detection remains challenging, with only two approaches approaching the ILAE Phase 3 benchmark (see Section 2.4). (Fine, 2025) reported the first automated tonic seizure detector, achieving 100% sensitivity at 0.023/h FAR with 14-second mean latency. (Dong et al., 2026) demonstrated that nocturnal monitoring reduces false positives substantially, attaining 71.6% sensitivity at 0.165/h using CNN-LSTM with attention mechanisms. By contrast, (Spahr et al., 2025) achieved 96% sensitivity for convulsive seizures but required a 30-model ensemble CNN to maintain <1/8 nights FAR.

ECG-based detection demonstrates a severe sensitivity-FAR trade-off that renders it clinically impractical for standalone deployment. (Reintjes et al., 2025) comprehensively evaluated three anomaly detection methods on the SeizeIT2 dataset (n=120, 856 seizures), revealing that high sensitivity and low FAR are mutually exclusive configurations. In the sensitivity-optimized setting, Matrix Profile achieved 98.16% sensitivity at 13.90 FA/h, while its FAR-optimized configuration dropped to 38.04% sensitivity at 1.91 FA/h. TimeVQVAE-AD showed a similar pattern: 96.43% sensitivity at 39.75 FA/h versus 61.09% sensitivity at 4.22 FA/h. Even the most conservative approach (MADRID FAR-optimized) achieved only 1.80% sensitivity while maintaining 0.05 FA/h. These false alarm rates remain orders of magnitude above the ILAE Phase 3 benchmark (see

Section 2.4), and no single configuration simultaneously achieves viable sensitivity with acceptable FAR[1].

A key advantage emerges from temporal context: nighttime monitoring achieves 1/61 nights false alarm rates versus 1/9 days during daytime. Physiological analysis reveals heart rate increases approximately 100 seconds before movement onset, offering a potential detection window. However, offline processing predominates across studies, with validation on embedded devices notably absent.

The detection scope remains restricted to convulsive seizures-tonic-clonic, tonic, and hypermotor variants. Non-motor seizures are largely undetectable with current non-EEG modalities. While deep learning advances now make automated tonic seizure detection feasible, bridging the gap between offline performance and real-time embedded deployment remains the primary barrier to clinical implementation.

6.3 Forecasting Performance

The most significant barrier to clinical translation is the responder problem: only a subset of patients achieve usable forecasting performance. When evaluated with rigorous leave-one-subject-out (LOSO) cross-validation, only 43.5% of patients achieve better-than-chance performance. This represents the most credible performance estimate to date. In contrast, studies employing patient-specific model tuning reported substantially higher success rates, but patient-specific optimization inflates performance compared to LOSO approaches, creating unrealistic expectations for real-world deployment where prospective tuning is infeasible.

Across successful cases, systems achieved mean warning times of approximately 30 minutes (ranging from 15 seconds to several hours) with AUC values spanning 0.74 to 0.97. The time spent in warning ranged from 14% to 47% depending on the alarm threshold. Circadian and multiday cycles emerged as the most reliable predictors, with 100% hourly forecasting success demonstrated using these patterns in one study. Conversely, wearable-only features failed at daily resolution, as found in multi-center research showing performance depended entirely on diary features rather than sensor data.

The main clinical barrier remains the inability to predict which patients will respond. Higher baseline seizure frequency correlates with poorer forecasting performance, and no validated biomarkers exist for patient selection. Until responder characteristics can be identified prospectively, deep learning forecasting systems cannot be deployed clinically, as clinicians cannot determine which patients will benefit from the intervention.

6.4 Methodological Trends

Algorithmic approaches have evolved substantially from 2013 to 2026. Early work relied on traditional machine learning (KNN, SVM, ANN) with handcrafted features extracted from cardiac signals. The period 2016-2019 saw a transition to heart rate variability-based deep learning with MLP and LSTM networks, though patient-specific tuning inflated apparent performance. From 2020-2022, hybrid architectures combining LSTM with convolutional layers emerged, alongside ensemble methods and the introduction of LOSO cross-validation. The current period (2023-2026) features attention-based architectures, matrix profile anomaly detection, and self-attentive autoencoders.

[1] In (Reintjes et al., 2025), "responders" refers to a subgroup definition—patients with >50 BPM heart rate increase during seizures—not a detection success rate.

Despite algorithmic sophistication, validation rigor has not kept pace. Only three of 13 studies employed prospective designs, and LOSO cross-validation remains rare. Several studies conducted validation in epilepsy monitoring units under controlled conditions that minimize motion artifacts and environmental noise, while others tested systems in home and ambulatory settings.

Metrics reporting lacks standardization, impeding cross-study comparison. False alarm rates appear as per-hour, per-day, or per-night metrics, or are omitted entirely. Patient-level success rates appear in only 6 of 13 studies. No widely adopted benchmark datasets exist beyond EPILEPSIAE.

Emerging technical directions include attention mechanisms for dynamic channel weighting, matrix profile approaches for unsupervised anomaly detection, and ensemble methods with tunable sensitivity thresholds. However, real-world deployment remains elusive: most algorithms process data on external servers rather than on the wearable device. Edge optimization has received minimal attention despite being vital for clinical translation, with only one study demonstrating real-time inference on smartwatch hardware.

6.5 Clinical Takeaways

Table 8 summarizes the best-performing detection systems. Only two approaches approach the ILAE Phase 3 benchmark (see Section 2.4).

Table 8: Highest Performing Detection Studies by Metric

Study	Seizure Type	Sens.	FAR	Latency	Validation
(Fine, 2025)	Tonic	100%	0.023/h	14 s	Phase 1
(Dong et al., 2026)	Nocturnal severe	71.6%	0.165/h	–	Prospective
(Spahr et al., 2025)	Convulsive	96%	<1/8d	26 s	Phase 2

For clinicians, (Dong et al., 2026)'s nocturnal monitoring system represents the most validated option for home use. (Fine, 2025)'s tonic seizure detector shows promise but requires Phase 2 validation. ECG-only approaches should be avoided due to unacceptable false alarm rates. For researchers, LOSO cross-validation should be adopted as the minimum standard for reporting performance, and standardized metrics including patient-level success rates must be reported to enable meaningful comparison across studies.

7 Conclusion

This systematic review examined thirteen studies on wearable seizure detection and forecasting using non-EEG biosignals (2013-2026). The evidence reveals a field at an inflection point: convulsive seizure detection has achieved technical maturity, yet major barriers prevent widespread clinical adoption.

7.1 The Translation Gap

The core finding is a disconnect between algorithmic performance and clinical deployability. While accelerometry-based systems achieve 90-100% sensitivity for convulsive seizures, only two of nine detection studies approach the ILAE Phase 3 benchmark for

validated devices (see Section 2.4). More importantly, ECG-based approaches - once considered promising due to preictal heart rate changes - exhibit an inherent trade-off where high sensitivity (96-98%) requires 13-40 FA/h, while low FAR (0.05-4.2/h) yields only 1.8-61% sensitivity. No single ECG configuration achieves viable sensitivity with acceptable false alarms.
This pattern suggests that **algorithmic refinement alone may be insufficient**. The field has exhausted substantial gains from architectural innovation (from KNN in 2013 to attention mechanisms in 2026) without commensurate improvements in clinical utility. The problem has shifted from "can we detect seizures?" to "can we deploy detectors that patients will tolerate?"
A fundamental scope limitation remains: non-EEG wearables can detect convulsive seizures but cannot identify non-motor seizures (focal aware, absence, or focal impaired awareness events), which constitute the majority of seizures for many patients. Even a perfectly optimized wearable system would remain incomplete without complementary EEG.

7.2 The Responder Problem

The most underappreciated barrier is patient heterogeneity. Leave-one-subject-out validation reveals that only 43.5% of patients achieve better-than-chance forecasting performance. For forecasting, patient success rates range from 43.5% to 100% across studies, with no reliable predictors of which patients will respond. Clinicians cannot prescribe wearable monitors with confidence that a given patient will benefit.
The responder problem has two implications rarely acknowledged. First, aggregated performance metrics are misleading. A system with 80% mean sensitivity may work perfectly for half of patients and fail completely for the other half. Second, personalized adaptation may be necessary rather than optional. Federated learning and patient-specific tuning, currently underexplored, warrant prioritization.

7.3 Validation Infrastructure Deficits

Methodological weaknesses consistently overestimate real-world performance. Several studies validate in epilepsy monitoring units, which minimize motion artifacts, data loss, and device displacement, while others tested systems in home and ambulatory settings with varying results. Also, monitoring durations vary widely across studies, biasing toward common seizure types, while rare events-precisely those most needing detection-remain underrepresented.
The field lacks standardized benchmark datasets. Without open datasets with annotated non-motor seizures, realistic artifact distributions, and multi-center diversity, algorithms cannot be compared meaningfully. The absence of ILAE Phase 3 evidence (large-scale prospective home validation) after more than a decade of research indicates that validation infrastructure, not algorithms, is the bottleneck.

7.4 The Explainability Gap

None of the thirteen studies in this review provide interpretable explanations for their detection or forecasting decisions. Most employed deep learning "black box" models - CNNs, LSTMs, ensemble methods - without exposing which features, time windows, or physiological patterns drive individual predictions. This omission is not merely academic.

Clinicians cannot recommend systems that cannot explain why an alarm was triggered, and regulatory approval increasingly requires interpretability for safety-critical medical AI.

Recent work demonstrates that explainability is achievable even for complex seizure models. (Gao et al., 2024) introduced the first interpretable and generalizable deep learning model for iEEG-based seizure prediction, using prototype learning to trace prediction origins to specific patient-level patterns. (Khan et al., 2024) combined fuzzy logic with deep learning to provide interpretable rules for EEG-based seizure prediction. (Vieira et al., 2023) demonstrated that XAI methods can simultaneously reduce the number of required EEG channels while maintaining performance-a dual benefit for wearable deployment.

For wearable systems specifically, (Halimeh et al., 2023) developed explainable AI for seizure logging, showing that data quality, patient age, and antiseizure medication considerably affect interpretability. (Germino et al., 2025) proposed a mixture-of-experts approach for explainable seizure detection using wearables, while (Shao et al., 2025) applied deconvolutional networks to visualize which features drive epilepsy detection decisions. Systematic reviews by (Gurmessa & Jimma, 2025) and (Zhao et al., 2023) confirm that interpretable methods can match black-box performance while providing clinical transparency.

The absence of explainability in the reviewed wearable studies represents a missed opportunity. Interpretability serves three clinical functions: building trust between clinicians, patients, and AI systems, enabling error analysis when false alarms occur, and supporting regulatory approval by providing safety assurances. Future wearable seizure monitors must prioritize interpretable architectures over opaque accuracy gains.

7.5 Research Priorities

Based on identified gaps, four priorities warrant immediate attention:

1. **Standardized reporting benchmarks.** Minimum requirements should include patient-level confusion matrices, latency distributions, and false alarm rates in consistent units. A community-agreed benchmark dataset with realistic artifact profiles would enable algorithm comparison on equal footing.

2. **Responder identification.** Research must identify baseline biomarkers (electroclinical, autonomic, or demographic) that predict which patients will achieve usable performance with wearable monitoring. Without this, clinical adoption remains trial-and-error.

3. **Interpretable architectures.** Future systems must employ explainable approaches-prototype learning, fuzzy deep learning, attention mechanisms with visualization, or mixture-of-experts-that provide clinical transparency without sacrificing detection performance. Black-box models that cannot explain their decisions face barriers to clinical trust and regulatory approval.

4. **Edge optimization and personalization.** Current algorithms predominantly run on external servers rather than on the wearable device itself. Real-world deployment requires lightweight architectures, model compression, and on-device learning for patient-specific adaptation without data centralization.

7.6 The Path Forward

The field has demonstrated that wearable seizure detection is **technically feasible**. The remaining challenge is **clinical implementation**. Success requires shifting focus from incremental accuracy improvements to solving the responder problem, reducing false alarms to tolerable levels, and validating systems in the environments where they will actually be used. Without this shift, wearable seizure monitoring risks becoming a technology that works in principle but cannot be prescribed in practice.

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